MA 5755: Data Analysis & Visualization in R/Python/SQL

Week 8: Regression, Regularization, and Model Selection

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Outline

1. Regression Setup
2. Worked Example
3. Optimization via Gradient Descent
4. Model Selection
5. Regularization

Problem Setup

We observe:

$$(x_i, y_i), \quad i = 1, \ldots, n$$

Assume:

$$y = f(x) + \epsilon$$

Goal:

$$\text{Estimate } f(x)$$

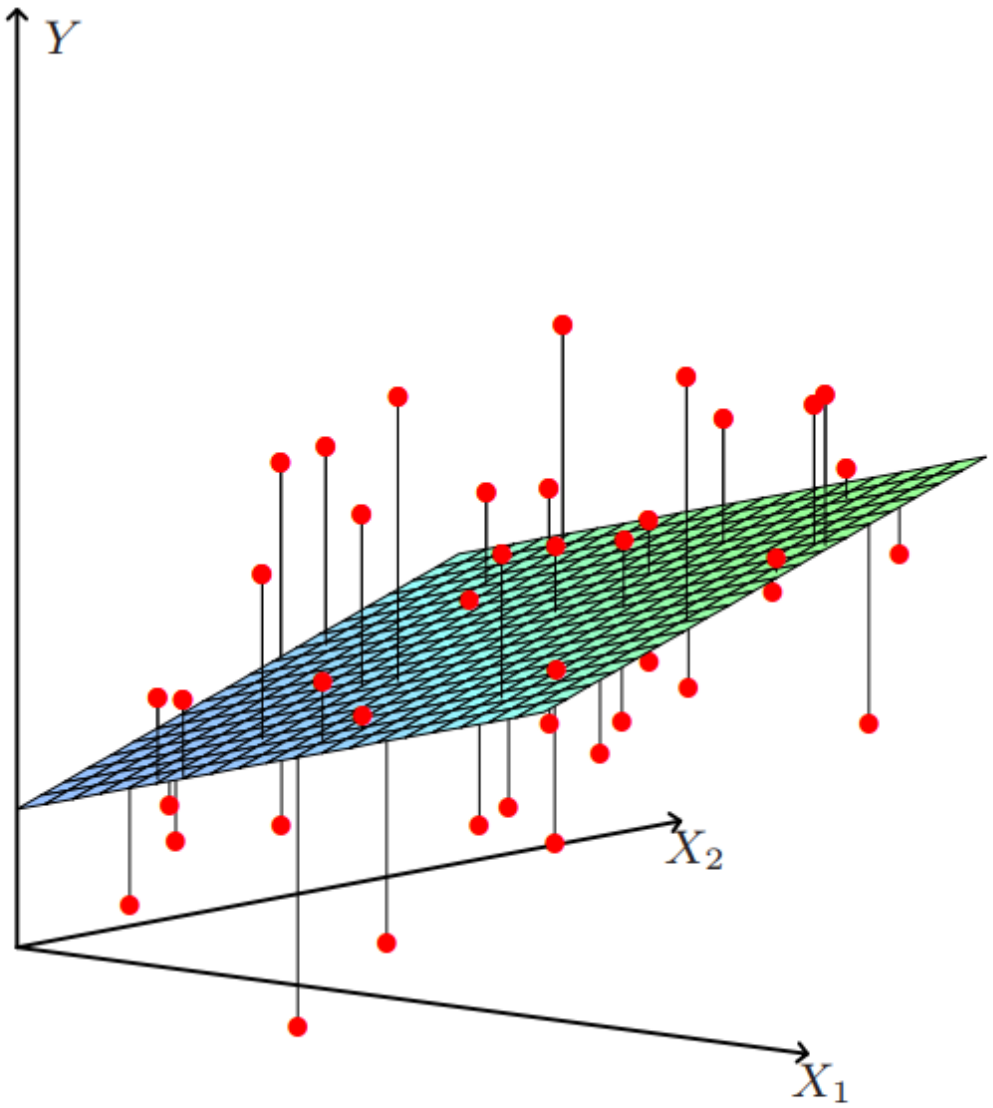
Restrict to linear class:

$$f(x) = \beta_0 + \sum_{j=1}^{p} \beta_j X_j$$

We solve:

$$\min_{\beta} \sum (y_i - \beta_0 + \sum_{j=1}^{p} (-1) \beta_j X_j)^2$$

Picture



Linear Models: Structure of Inputs

Key point: **linear in parameters** $\{\beta_j\}$, not necessarily in original inputs.

Sources of predictors X_j (other than raw quantitaive inputs X_j)

- **Transformations:**

$$X_j = \log x, \ \sqrt{x}, \ x^2$$

- **Basis expansions (polynomial):**

$$X_1 = x, \quad X_2 = x^2, \quad X_3 = x^3$$

- **Categorical variables (dummy coding):**

$$X_j = \mathbf{1}(G = j), \quad j = 1, \ldots, K, \ \sum_{j=1}^{K} \beta_j X_j \Rightarrow \text{level-specific constants}$$

- **Interactions:**

$$X_3 = X_1 \cdot X_2$$

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Worked Example: Dataset

Data:

$$(1,2), \quad (2,3), \quad (3,5)$$

Model:

$$y = \beta_0 + \beta_1 x$$

Matrix form:

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad y = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$$

Goal:

$$\min_{\beta} \|y - X\beta\|^2$$

Normal Equations (Step-by-Step)

We compute:

$$X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \quad X^T y = \begin{bmatrix} 10 \\ 23 \end{bmatrix}$$

Normal equation:

$$(X^T X)\beta = X^T y$$

System:

$$\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 10 \\ 23 \end{bmatrix}$$

Solve:

$$\beta_0 = \frac{4}{3}, \quad \beta_1 = \frac{3}{2}$$

Model:

$$\hat{y} = \frac{4}{3} + \frac{3}{2}x$$

Residuals in Example

Predictions:

$$\hat{y}_1 = 2.83, \quad \hat{y}_2 = 4.33, \quad \hat{y}_3 = 5.83$$

Residuals:

$$r = y - \hat{y}$$

$$r = (-0.83, -1.33, -0.83)$$

Observation:

- Residuals not symmetric $\rightarrow$ possible model mismatch

Least Squares as Projection in $\mathbb{R}^n$

Model:

$$y = X\beta, \quad X = [x_0, x_1, \ldots, x_p]$$

Columns of X span a subspace $\mathcal{S} \subset \mathbb{R}^n$.

Key condition:

$$X^T(y - X\hat{\beta}) = 0$$

Interpretation:

- Residual $r = y - \hat{y}$ is orthogonal to $\mathcal{S}$
- $\hat{y} = X\hat{\beta}$ is the **orthogonal projection** of y onto $\mathcal{S}$

Projection operator (hat matrix):

$$H = X(X^TX)^{-1}X^T, \quad \hat{y} = Hy$$

Rank Deficiency and Non-Uniqueness

If columns of X are linearly dependent:

$$\text{rank}(X) < p + 1 \quad \Rightarrow \quad X^T X \text{ singular}$$

Consequences:

- $\hat{\beta}$ is **not unique**, multiple coefficient vectors give same fit

However $\hat{y} = X\hat{\beta}$ is still unique.
Reason:

- Projection onto subspace $\mathcal{S}$ is unique
- Only representation in terms of columns is non-unique

Practical causes:

- Perfect multicollinearity (e.g., $x_2 = 3x_1$), Redundant dummy variables, High-dimensional setting ($p > n$)

Statistical Properties of Least Squares

Assumptions: $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$. Then, the distribution of estimator:

$$\hat{\beta} \sim \mathcal{N}\left(\beta, (X^T X)^{-1}\sigma^2\right)$$

Variance estimate:

$$\hat{\sigma}^2 = \frac{1}{n-p-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Properties: $\mathbb{E}[\hat{\beta}] = \beta$ (unbiased), $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$.

Additional results: $(n-p-1)\hat{\sigma}^2 \sim \sigma^2 \chi^2_{n-p-1}$

Use confidence intervals for β_j and perform hypothesis testing.

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Gradient Derivation (Step-by-Step)

Loss: $L(\beta) = \frac{1}{n}(y - X\beta)^T(y - X\beta)$

Expand:

$$= \frac{1}{n}(y^T y - 2\beta^T X^T y + \beta^T X^T X\beta)$$

Differentiate:

$$\nabla L(\beta) = -\frac{2}{n}X^T y + \frac{2}{n}X^T X\beta$$

$$= -\frac{2}{n}X^T(y - X\beta)$$

Update rule:

$$\beta_{t+1} = \beta_t - \eta\nabla L(\beta_t)$$

Substitute gradient:

$$\beta_{t+1} = \beta_t + \frac{2\eta}{n}X^T(y - X\beta_t)$$

Gradient Descent Update and Interpretation

Rewrite:

$$\beta_{t+1} = \beta_t + \frac{2\eta}{n} X^T r_t, \quad r_t = y - X\beta_t$$

Interpretation:

- Update driven by correlation between features and residuals
- If $X_j^T r_t \neq 0$, coefficient β_j is adjusted

Connection to normal equations:

$$X^T(y - X\beta^*) = 0$$

- Gradient descent seeks β such that residual is orthogonal to column space

Convergence:

- Depends on eigenvalues of $X^T X$
- Poor conditioning $\Rightarrow$ slow convergence (zig-zag path)

Bias–Variance Decomposition

Prediction error at x:

$$\mathbb{E}\big[(y - \hat{f}(x))^2\big] = \mathbb{E}\big[(f(x) + \epsilon - \hat{f}(x))^2\big]$$

Decomposition:

$$= (\mathbb{E}[\hat{f}(x)] - f(x))^2 + \mathbb{E}\big[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2\big] + \sigma^2$$

Components:

- **Bias**^{2}: $(\mathbb{E}[\hat{f}(x)] - f(x))^2$ (approximation error)
- **Variance**: $\mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]$ (estimation instability)
- **Noise**: σ^2

Implication:

$$\text{Test Error} = \text{Bias}^2 + \text{Variance} + \sigma^2$$

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Why Not Least Squares Alone?

Least squares:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Two fundamental issues:

1. **Prediction Accuracy**
   - Low bias but potentially high variance
   - Sensitive to noise and multicollinearity
2. **Interpretation**
   - Large $p \to$ difficult to interpret
   - Desire for **parsimonious models**

Idea:

Select a subset of variables to balance bias and variance

Subset Selection Framework

Goal:

$$\min_{\beta} \|y - X_S \beta_S\|^2$$

where $S \subset \{1, \ldots, p\}$ is a subset of variables.
Key idea:

- Fit least squares on selected variables only
- Eliminate others (set coefficients to zero)

Produces:

- Sequence of models indexed by subset size $k = |S|$
- Increasing $k \to$ decreasing training error

Key problem:

How to choose optimal subset size k?

Best Subset Selection

For each $k = 0, 1, \ldots, p$:

$$\text{Find } S_k = \arg\min_{|S|=k} \|y - X_S \beta_S\|^2$$

Properties:

- Searches all 2^p subsets
- Returns best model for each size k

Important:

- Training RSS decreases with k
- Cannot choose k using training error alone

Tradeoff:

- Large $k \to$ low bias, high variance
- Small $k \to$ high bias, low variance

Limitation:

Computationally infeasible for large p

Stepwise Methods

Forward Stepwise

- Start with intercept
- Add variable that most reduces RSS
- Produces nested sequence

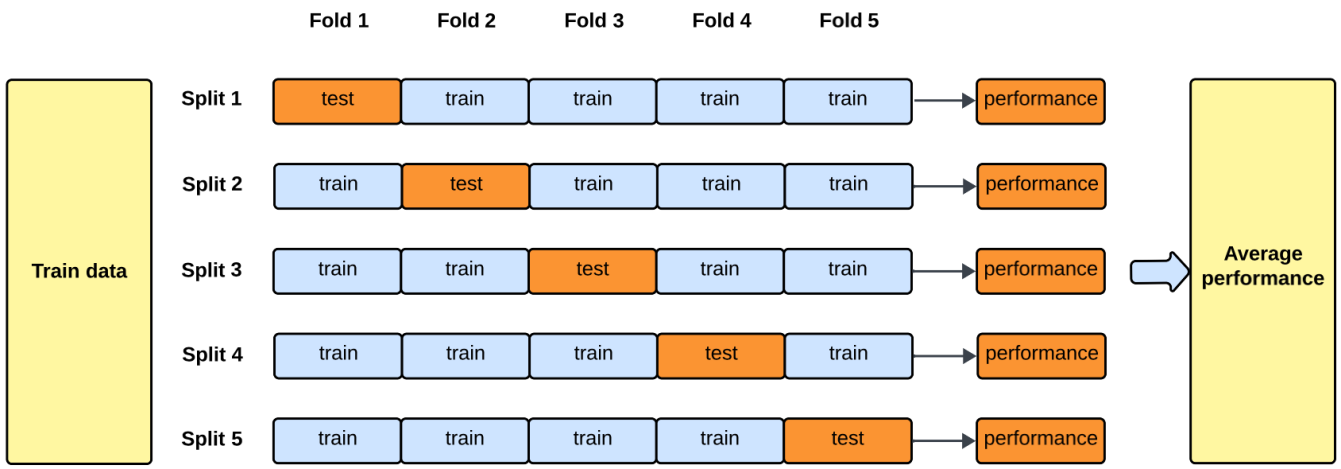
Backward Stepwise

- Start with full model
- Remove least useful variable
- Requires $n > p$

Forward Stagewise

- Adds very small increments to coefficients
- Extremely conservative updates
- Useful in high-dimensional settings

Cross-Validation



Split data into k folds, train on $k-1$ folds, validate on the remaining fold, and find the average CV error. Compute CV error for each subset size k. Select $\lambda^* = \arg\min \text{CV error}$.

Practical refinement: Choose the simplest model within one standard error of minimum

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From Subset Selection to Regularization

Subset selection can reduce bias, but often has high variance.

Issue: Model selection over 2^p subsets $\Rightarrow$ unstable

Idea: **Shrinkage (continuous alternative)**

$$\min_{\beta} \|y - X\beta\|^2 + \lambda\Omega(\beta)$$

Key effects:

- Shrinks coefficients toward zero
- Reduces variance without hard selection
- Produces a continuum of models (controlled by λ)

Goal:

Improve prediction by bias–variance tradeoff

Ridge vs LASSO: Key Difference

Ridge (L2 penalty)

$$\Omega(\beta) = \sum \beta_j^2$$

- Shrinks coefficients continuously
- No exact zeros

LASSO (L1 penalty)

$$\Omega(\beta) = \sum |\beta_j|$$

- Produces sparse solutions
- Performs variable selection

Key contrast: Ridge: shrinkage LASSO: shrinkage + selection

Ridge vs lasso in picture

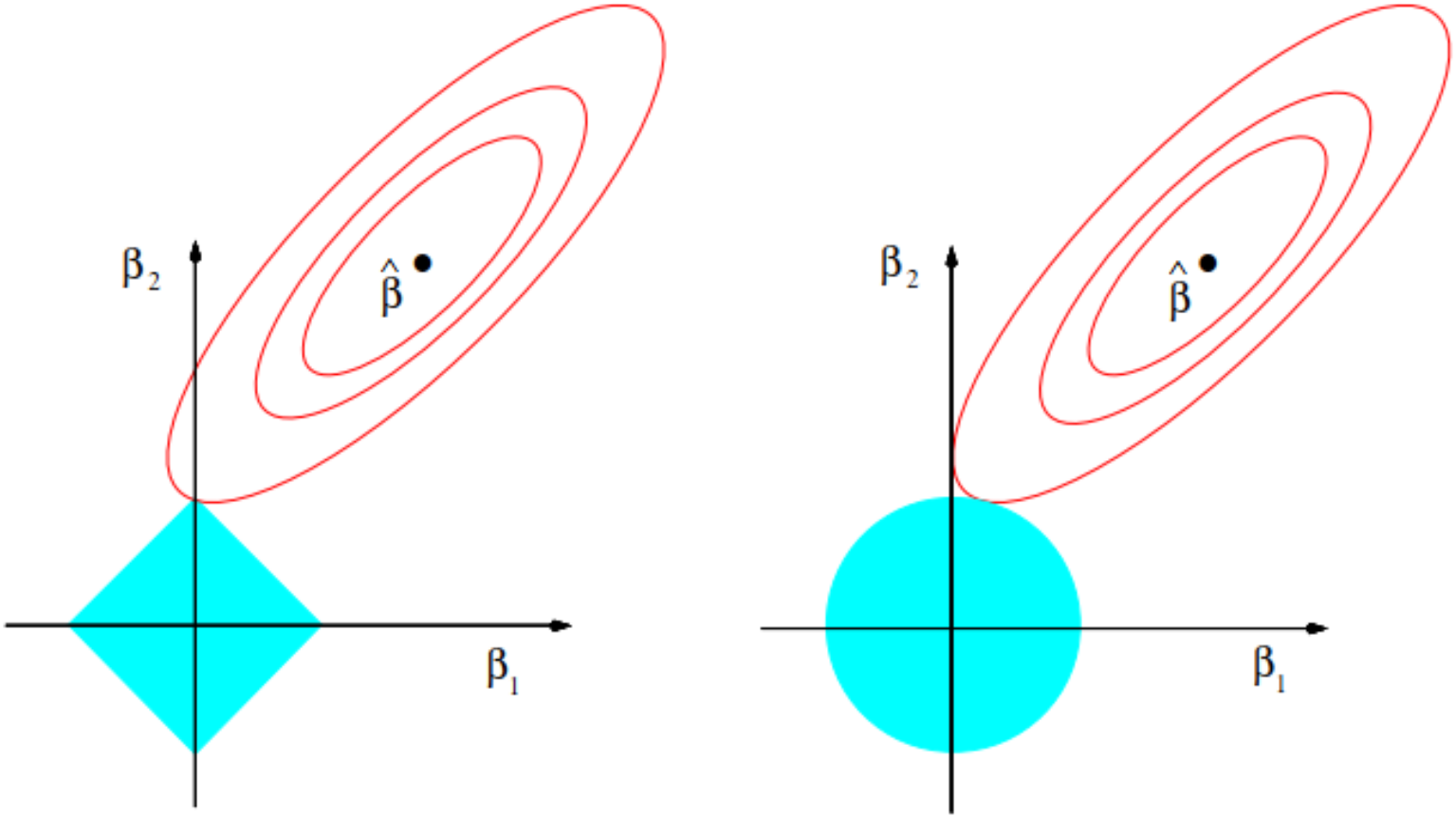


FIGURE 3.11. *Estimation picture for the lasso (left) and ridge regression (right). Shown are contours of the error and constraint functions. The solid blue areas are the constraint regions* $|\beta_1| + |\beta_2| \leq t$ *and* $\beta_1^2 + \beta_2^2 \leq t^2$*, respectively, while the red ellipses are the contours of the least squares error function.*

Ridge Regression: Formulation

Objective:

$$\hat{\beta}_{ridge} = \arg\min_{\beta} \|y - X\beta\|^2 + \lambda \sum \beta_j^2$$

Equivalent constrained form:

$$\min_{\beta} \|y - X\beta\|^2 \quad \text{s.t.} \quad \sum \beta_j^2 \leq t$$

Closed-form solution:

$$\hat{\beta}_{ridge} = (X^T X + \lambda I)^{-1} X^T y$$

Effect:

- Adds $\lambda I \rightarrow$ stabilizes inversion
- Handles multicollinearity

Ridge Regression: Geometric and Spectral View

Geometric view:

- Constraint: $\sum \beta_j^2 \leq t$ (sphere)
- Solution = intersection with RSS contours

SVD: $X = UDV^T$

Fitted values:

$$X\hat{\beta}_{ridge} = \sum_{j=1}^{p} u_j \frac{d_j^2}{d_j^2 + \lambda} u_j^T y$$

Interpretation:

- Shrinks components along eigen-directions
- Stronger shrinkage for small d_j

Key effect: Reduces variance in poorly determined directions

Profile plot: Ridge

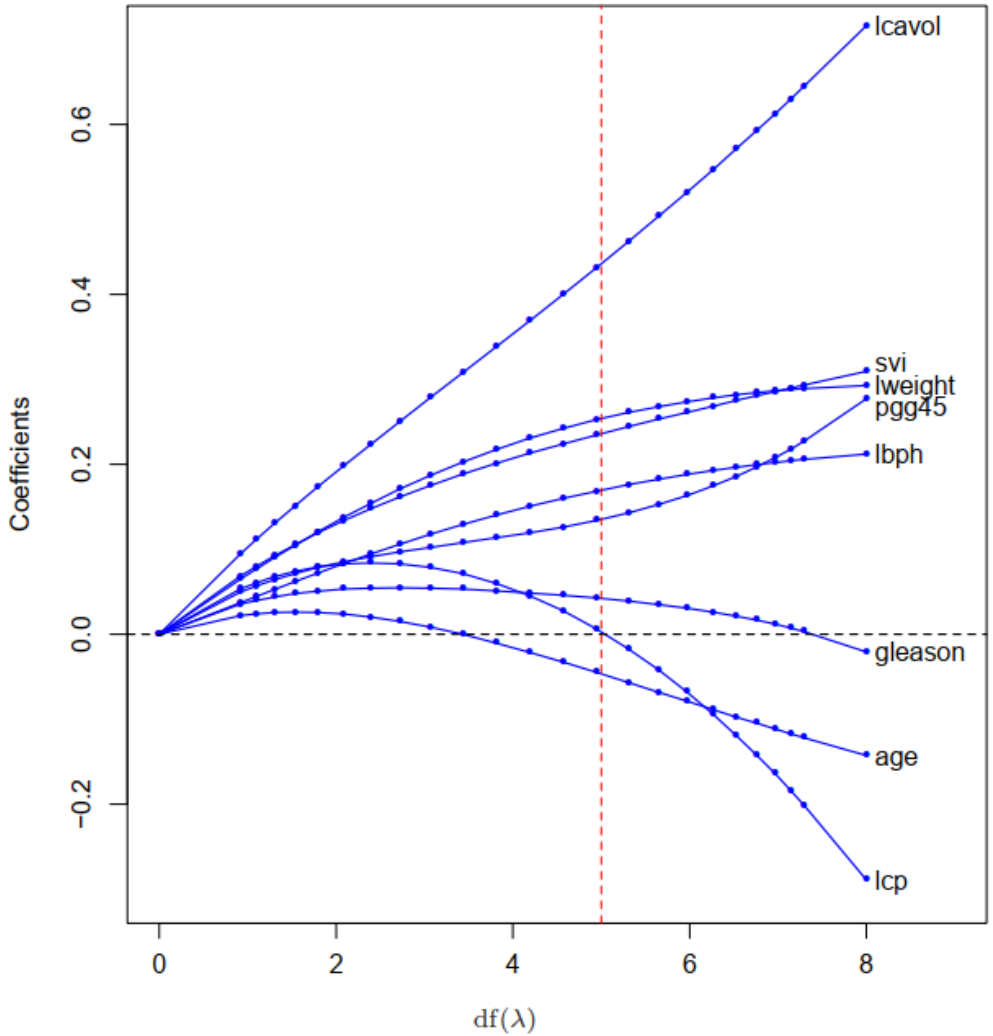


FIGURE 3.8. *Profiles of ridge coefficients for the prostate cancer example, as the tuning parameter λ is varied. Coefficients are plotted versus* $\mathrm{df}(\lambda)$, *the effective*

LASSO: Formulation and Sparsity

Objective:

$$\hat{\beta}_{lasso} = \arg\min_{\beta} \|y - X\beta\|^2 + \lambda \sum |\beta_j|$$

Equivalent constraint:

$$\sum |\beta_j| \leq t$$

Key property:

- Non-differentiable at $\beta_j = 0$
- Leads to exact zeros in coefficients

Result:

$$\text{Sparse model} \Rightarrow \text{automatic variable selection}$$

Lasso: Profile plots

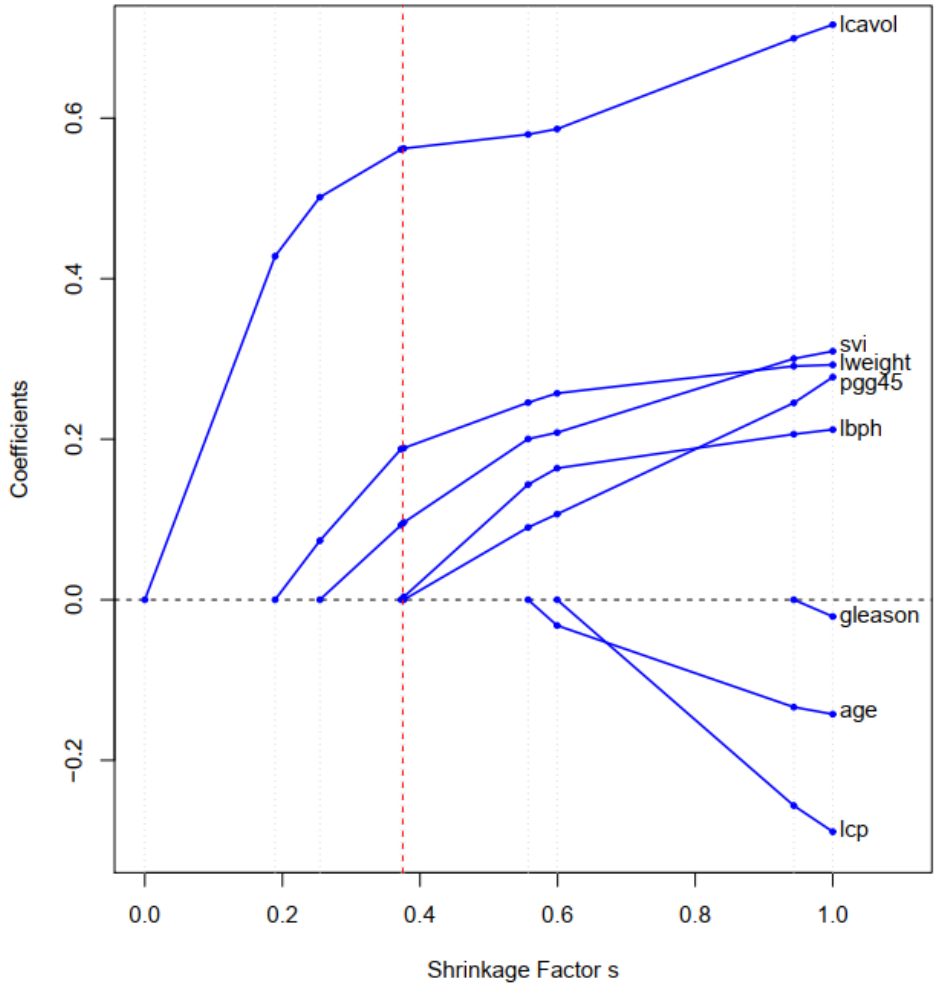


FIGURE 3.10. *Profiles of lasso coefficients, as the tuning parameter t is varied. Coefficients are plotted versus $s = t/\sum_1^p |\hat{\beta}_j|$. A vertical line is drawn at $s = 0.36$, the value chosen by cross-validation. Compare Figure 3.8 on page 65; the lasso*

LASSO: Geometry and Behavior

Constraint region:

- L2 (ridge): sphere
- L1 (lasso): diamond

Key insight:

- Corners of L1 constraint $\rightarrow$ coefficients exactly zero

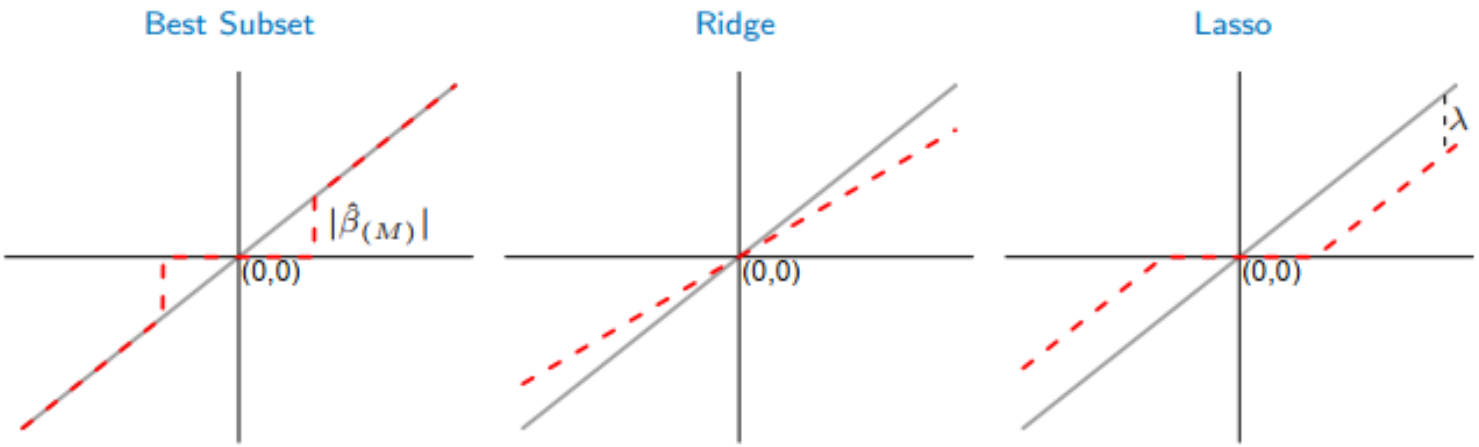
Comparison:

- Ridge: distributes weight across correlated variables
- LASSO: selects one, discards others

Tradeoff:

- Ridge $\rightarrow$ lower variance
- LASSO $\rightarrow$ interpretable sparse model

Estimates

TABLE 3.4. *Estimators of β_j in the case of orthonormal columns of* $\mathbf{X}$. *M and λ are constants chosen by the corresponding techniques;* sign *denotes the sign of its argument (± 1), and x_+ denotes "positive part" of x. Below the table, estimators are shown by broken red lines. The 45° line in gray shows the unrestricted estimate for reference.*

Estimator	Formula
Best subset (size M)	$\hat{\beta}_j \cdot I(\lvert\hat{\beta}_j\rvert \geq \lvert\hat{\beta}_{(M)}\rvert)$
Ridge	$\hat{\beta}_j/(1+\lambda)$
Lasso	$\text{sign}(\hat{\beta}_j)(\lvert\hat{\beta}_j\rvert - \lambda)_+$